

Assignment : 2

Section A

1. What does CDMA stand for in the context of Link Layer's Medium Access control?

→ CDMA stands for Code Division Multiple Access. It's a Method used in wireless communication to efficiently share the medium. It allows multiple devices to transmit data simultaneously by using unique codes.

2. Name one disadvantage of the ALOHA multiple access protocol?

→ Disadvantage of Aloha Multiple access protocol is that it can lead to collisions when multiple devices try to transmit data at the same time.

3. What is the primary purpose of Link Layer addressing?

→ The primary purpose of Link Layer addressing in computer networks is to uniquely identify devices on the same local network. It is responsible for the node-to-node

delivery of data & helps to ensure that data is delivered to the correct destination.

4. What is the maximum length of a standard Ethernet frame?

→ The maximum length of a standard Ethernet frame is 1518 bytes, which includes the Ethernet header and trailer. The size limitation helps ensure efficient data transmission and prevents network congestion. It's an important aspect of network communication protocols.

5. Define "spanning tree" in the context of network topology?

→ 1. Spanning Tree Protocol (STP) is a layer 2 network protocol used to prevent looping within a network topology.

2. STP was created to avoid the problems that arise when computer exchange data on LAN (Local Area Network) that contains redundant paths.

3. If the flow of traffic is not carefully monitored and controlled, the data can be caught in a loop that circles around a network segment, affecting performance & bringing traffic to a near halt.

SECTION: B

1. Compare & contrast CDMA and ALOHA as multiple access protocols, highlight their key differences

→ CDMA (Code Division Multiple Access) allows multiple users to share the same frequency band simultaneously by assigning a unique code to each other. This enables simultaneous transmission and reception of data, resulting in used capacity and improved efficiency. ALOHA, on the other hand, is a random access protocol where multiple users contend for access to the shared channel. When a user wants to transmit, they randomly choose a time slot and send their data. If there is a collision, the data is retransmitted after a random backoff period.

The key difference b/w CDMA and ALOHA is the way they handle multiple users.

CDMA uses the code division technique to allow simultaneous transmission, while ALOHA uses a random access approach where users contend for access to the channel.

1. Efficiency :- CDMA is known for its high capacity and efficient use of the available spectrum, as it allows multiple users to transmit simultaneously. whereas, ALOHA may experience lower efficiency due to collision and retransmission.
2. Synchronization :- CDMA requires users to be synchronized with the same spreading code. In ALOHA synchronization is not necessary as users contend for access randomly.
3. Scalability :- CDMA is highly scalable and can accommodate a large number of users. ALOHA may face scalability challenges in crowded networks due to increased collision as No. of users increases.
4. Collision Resolution :- CDMA ~~has~~ a technique called "Spread Spectrum" to minimize collision by assigning unique code to each user. In contrast, ALOHA relies on Random access which can result in collisions when multiple users transmit simultaneously.

2. Explain the concept of Link Layer Switching and how it improves network performance.

→ Link Layer Switching is the concept in networking where data packets are forwarded at the link layer of the network protocol stack. It involves the use of switches to connect devices within a Local Area Network (LAN).

When a switch receives a data packet, it examines the destination MAC address in the packet's header and determines the appropriate outgoing port to forward the packet. This process is known as MAC address learning.

Link Layer Switching improves network performance in several ways:→

1. **Reduced collisions:**→ Switches create separate collision domains for each port, reducing collisions that can occur in shared media networks like Ethernet hubs. This improves overall network efficiency and reduces data packet loss.
2. **Increased bandwidth:**→ By forwarding data packets only to the desired destination, switches effectively utilize the available network bandwidth. This reduces congestion and improves network performance.
3. **Enhanced security:**→ Switches isolate traffic between different devices connected to the network, preventing unauthorized access to data. This improves network security.

and reduces the risk of data breaches.

4. Improved scalability :- Switches allow for the creation of larger and more complex networks by connecting multiple devices. They can handle a higher volume of network traffic & provide better scalability compared to traditional hubs.

Overall, Link Layer Switching optimizes network performance by reducing collisions, using bandwidth utilization, enhancing security & improving scalability.

3. Describe the channel Allocation problem in wireless communication. What are the main challenges associated with channel allocation in wireless network?

→ Channel allocation is a process in which a single channel is divided and allotted to multiple users in order to carry out user specific tasks. There are user's quantity may vary every time the process takes place. If there are N number of users and channels, each user is assigned one position. If no. of users are small and don't vary at times, then frequency

Division Multiplexing can be used as it is a simple and efficient channel bandwidth allocating technique.

Channel allocation problems

↓

Static channel

Allocation in

LAN's and MAN's

↓

Dynamic

channel allocation

The Main challenges associated with channel allocation in wireless network include:-

1) Limited Spectrum:- The availability of spectrum is limited and needs to be efficiently allocated among various users. This becomes challenging as No. of user and their demands for bandwidth rises.

2) Interference:- Wireless Net. are susceptible to interference from other devices operating in same frequency range. Allocating channel in a way that minimizes interference and maximizes Net. performance is a complex task.

3) Dynamic environment:- Wireless Net. operate in dynamic environment where No. of user and their spatial distribution can change frequently. This makes it challenging to allocate channels effectively and adaptively.

4) Co existence of Technologies:- Wireless Net. often supports multiple technologies like

WiFi, Bluetooth, etc. Coordinating the allocation of channels across different tech. without causing interference or performance degradation is a significant challenge.

5. **Scalability:** → As No. of devices and users ↑, channel allocation algorithms need to be Scalable to handle the rising complexity and demands of network.

SECTION: C

1. Explain Principles of Ethernet as a Multiple access protocol. Discuss its collision detection mechanism and how it handles collision?

➤ Ethernet is a multiple access protocol that allows multiple devices to share the same network medium. It uses principle called Carrier Sense Multiple Access with collision detection (CSMA/CD) to manage collisions. They may occur when multiple devices try to transmit data simultaneously.

Ethernet's Collision detection Mechanism Works :->

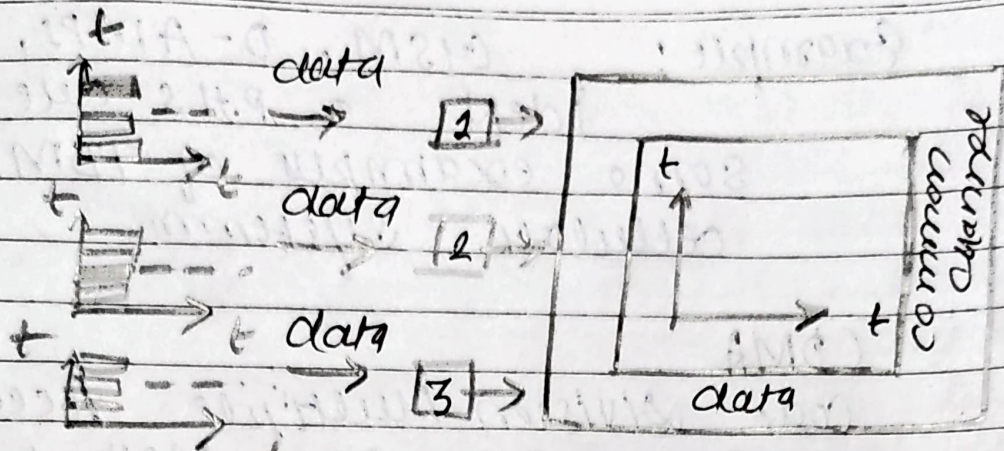
1. **Carrier Sense** :-> Before transmitting data, devices using Ethernet listen to the Network to check if it is idle. If the Network is busy, they wait for a random amount of time before attempting to transmit again.
2. **Multiple access** :-> Multiple devices can access the Net. Medium. Simultaneously. However, collision can occur if 2 or more devices transmit data at same time.
3. **Collision detection** :-> While transmitting devices continuously monitor the Network to detect collision. Collision happens when multiple devices transmit data and their signals overlap.
4. **Exponential Backoff** :-> If multiple collisions occur consecutively, backoff time T_{SE} exponentially to avoid repeated collision and give more time for Network to clear.
5. **Backoff Algorithm** :-> After the signal jam, colliding devices wait for a random amount of time before reattempting transmission. This helps to reduce chances of another occurring collisions.

It handles collision by immediately stop transmitting and send a jam signal to notify all other devices on the network about collision. After that each colliding device waits for a random amount of time before reattempting transmission using a backoff algorithm. This helps to reduce the chances of another collision and ensures fair access to the network medium.

2. Describe FDMA, TDMA & CDMA with examples.

Frequency Division Multiple Access is a multiple access protocol that allows multiple users to share the same frequency band by assigning diff. frequencies to each user.

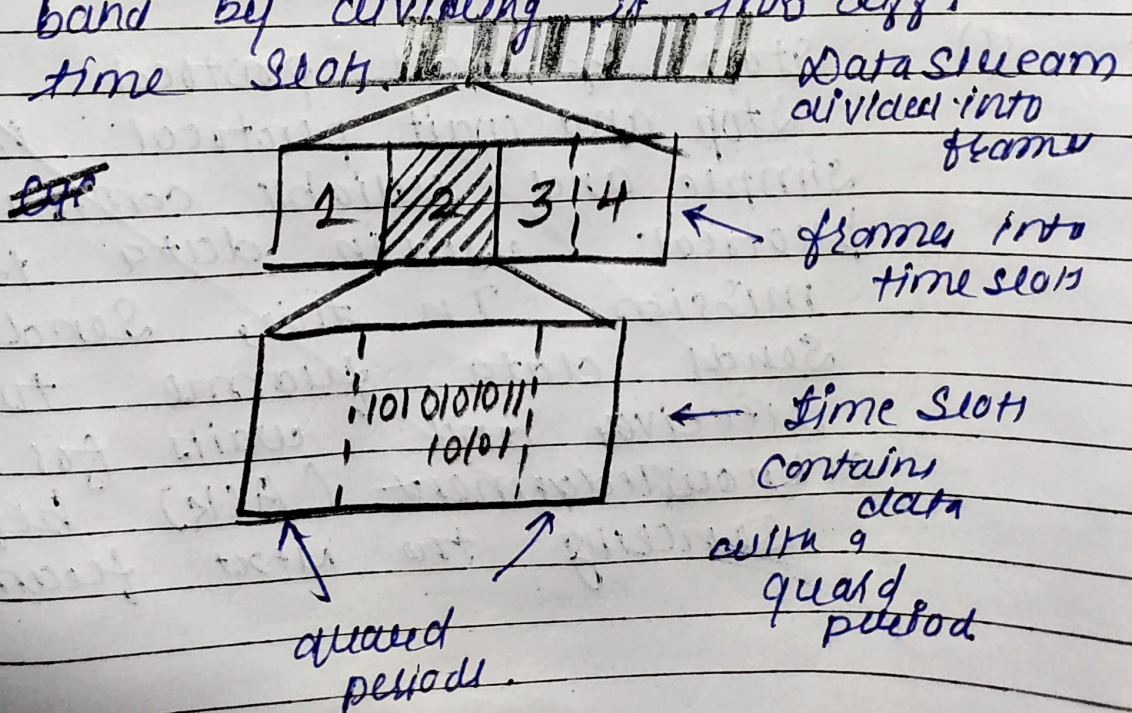
It is a type of channelization protocol. Its bandwidth is divided into various freq. bands. Each station is allocated a band to send data. It is reserved for the particular station for all the time.



Walkie Talkies and Mobile Networks for cloud user groups often use FDMA. Another example of FDMA is AM or FM radio broadcasting, where each station has its own channel.

TDMA

Time Division Multiple Access, is Multiple access protocol that allows Multiple users to share the same frequency band by dividing it into diff. time slots.



Example: GSM, D-AMPS, PDC
 IS-54 & IS-136 are
 some examples of TDMA
 cellular systems.

CDMA

Code Division Multiple Access is
 a channel access method used
 by several radio communication
 techniques. CDMA allows
 several users to share a
 band of frequencies without
 interference. Also, the user
 it is used as an access
 method in many mobile phones.

3. Write short note on:-

- (1) Stop & wait protocol
 Stop and wait protocol is a
 simple and reliable communication
 protocol used in data transfer
 missions. In this, Sender
 sends data frame to the
 receiver and waits for an
 acknowledgement (ACK) before
 sending the next frame.

The data stop and wait protocol is a flow control protocol is one of the services of the data link layer. Whenever the data that sender wants to send, he sends to receiver. After sending, he stops and waits until he receives acknowledgement from the receiver.

(II) Go-back-N-ARQ Protocol

Go back-N ARQ (Automatic Repeat Request) is a data link layer protocol that is used for data flow control purposes. It's a sliding window protocol in which multiple frames are sent from sender to receiver at once. The number of frames that are sent at once depends on the size of window that is taken.

(III) Selective Repeat

Selective Repeat Protocol (SRP) is a type of error control protocol we use in CN to ensure the reliable delivery of data packets. Additionally we use it in conjunction with the transmission control protocol (TCP) to ensure receiver receives data transmitted over the Net. without errors.